SigmaStar Tool Instructions

1. Sstar System Tool Description

Software developers must use Debug Tool hardware tools and Sstar System Tool software tools to access SigmaStar chip registers. Debug Tool hardware tool, as shown.

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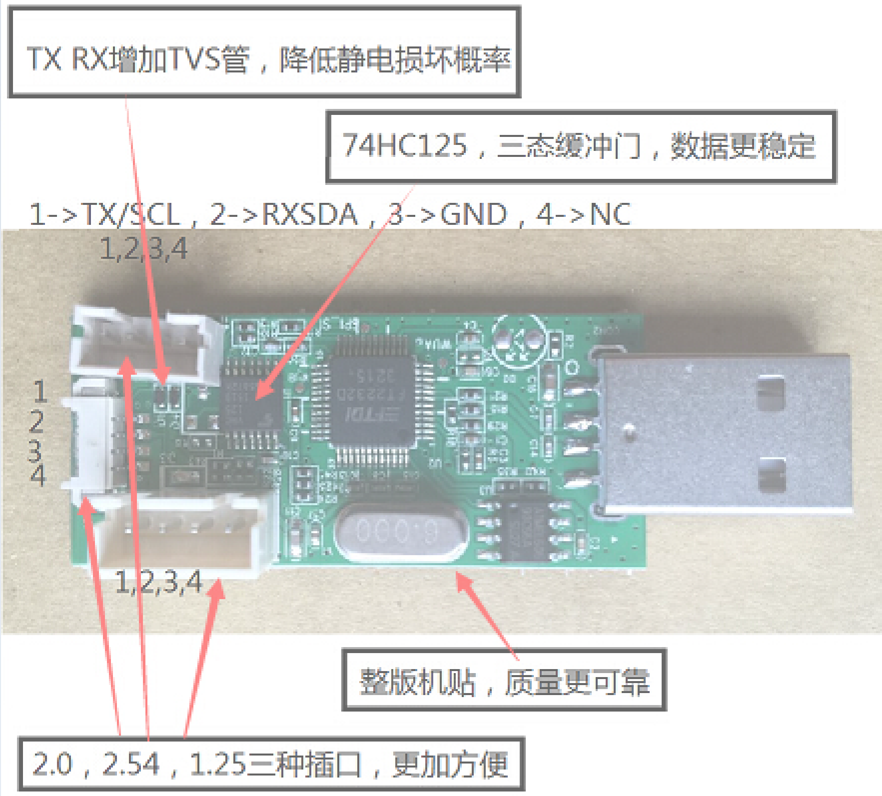


Picture 1-1

Use a USB extension cable to connect to the PC, and install the driver "debug tool driver for win7".

Open the Securt CRT tool. There are currently two ways to stop the serial port:

1. Enter uboot, type the command debug, and type Enter `关闭串口终端`, as shown in the following figure:

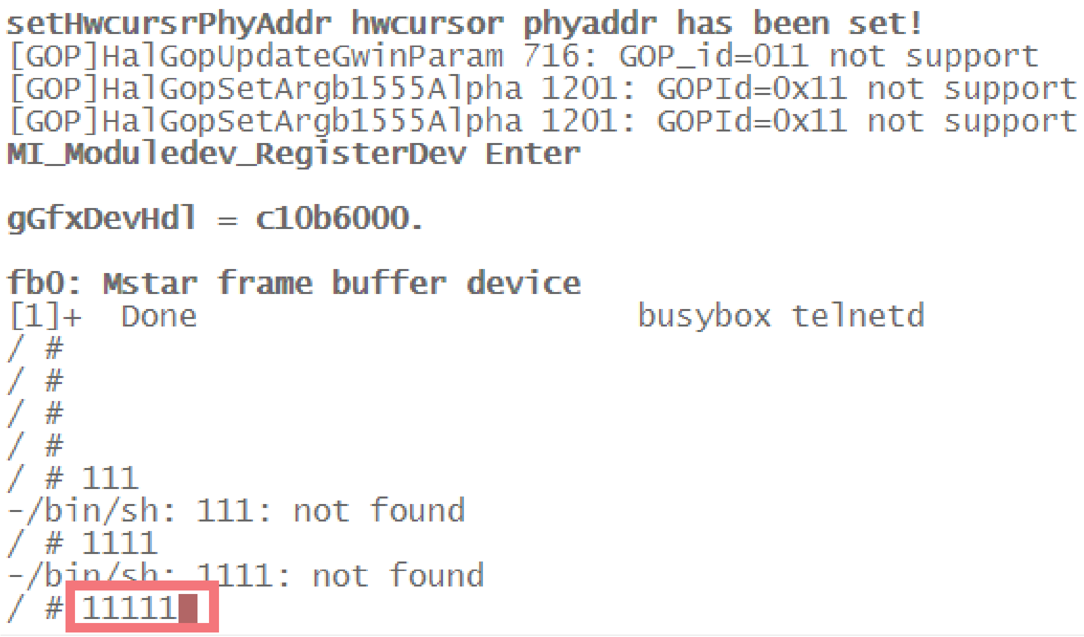
   ```
   SigmaStar #
   SigmaStar #
   SigmaStar #
   SigmaStar # debug

   debug mode on, cmdline is disabled

   SigmaStar #
   ```

   Figure 1-2

2. Enter the system, enter 11111 (5 1s), stop the serial port, `关闭串口终端`, as shown in the following figure:



```
setHwcursrPhyAddr hwcursor phyaddr has been set!
[GOP]HalGopUpdateGwinParam 716: GOP_id=011 not support
[GOP]HalGopSetArgb1555Alpha 1201: GOPId=0x11 not support
[GOP]HalGopSetArgb1555Alpha 1201: GOPId=0x11 not support
MI_Moduledev_RegisterDev Enter

gGfxDevHdl = c10b6000.

fb0: Mstar frame buffer device
[1]+  Done                       busybox telnetd
/ #
/ #
/ #
/ #
/ # 111
-/bin/sh: 111: not found
/ # 1111
-/bin/sh: 1111: not found
/ # 11111
```

Figure 1-3

Open the interface as shown in Figure 1-4, confirm the interface `chip /interface/Slave addr` , select as shown in Figure 1-4 below, and then click the button, the interface shown in Figure 1-5 will appear

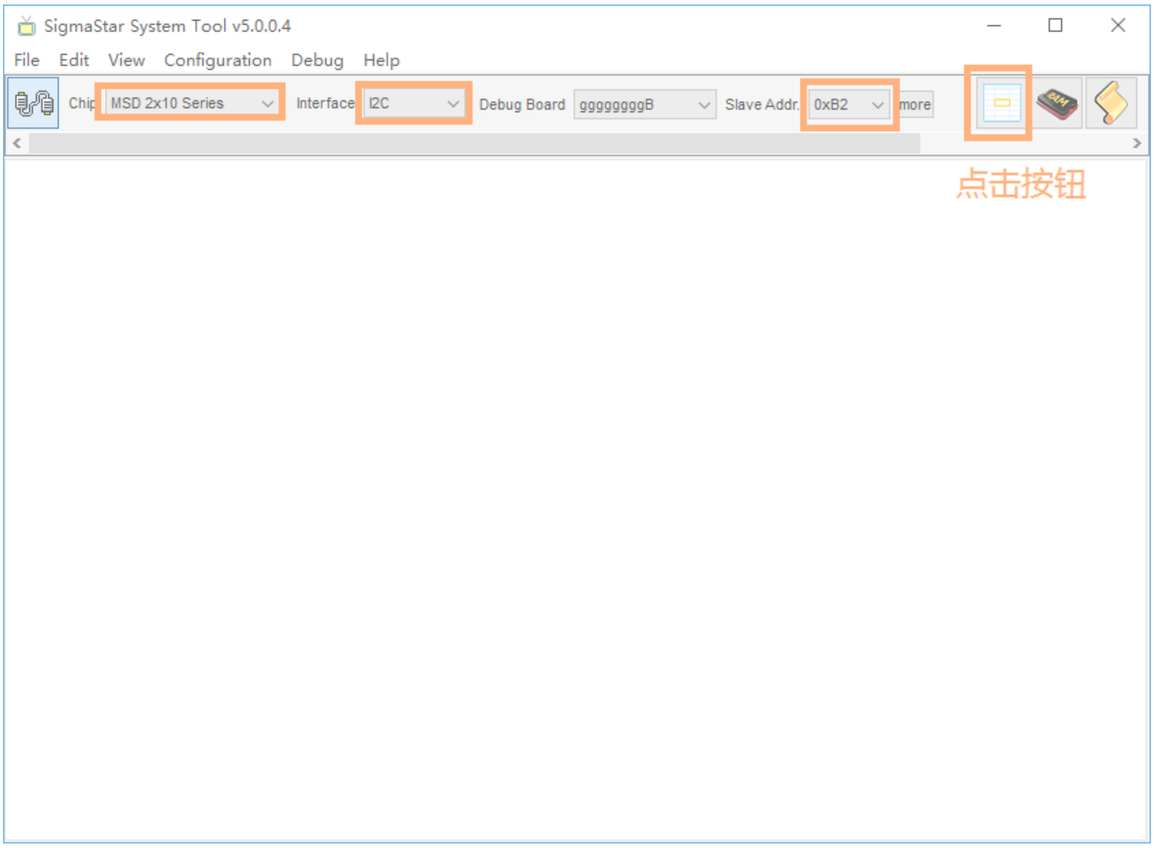


Figure 1-4

如下图1-5界面中，选择bank，范围在1002-1016区间，选择MIU，然后点击Read Bank，若能正常读到数据，则表示Sstar tool和芯片连接正常。

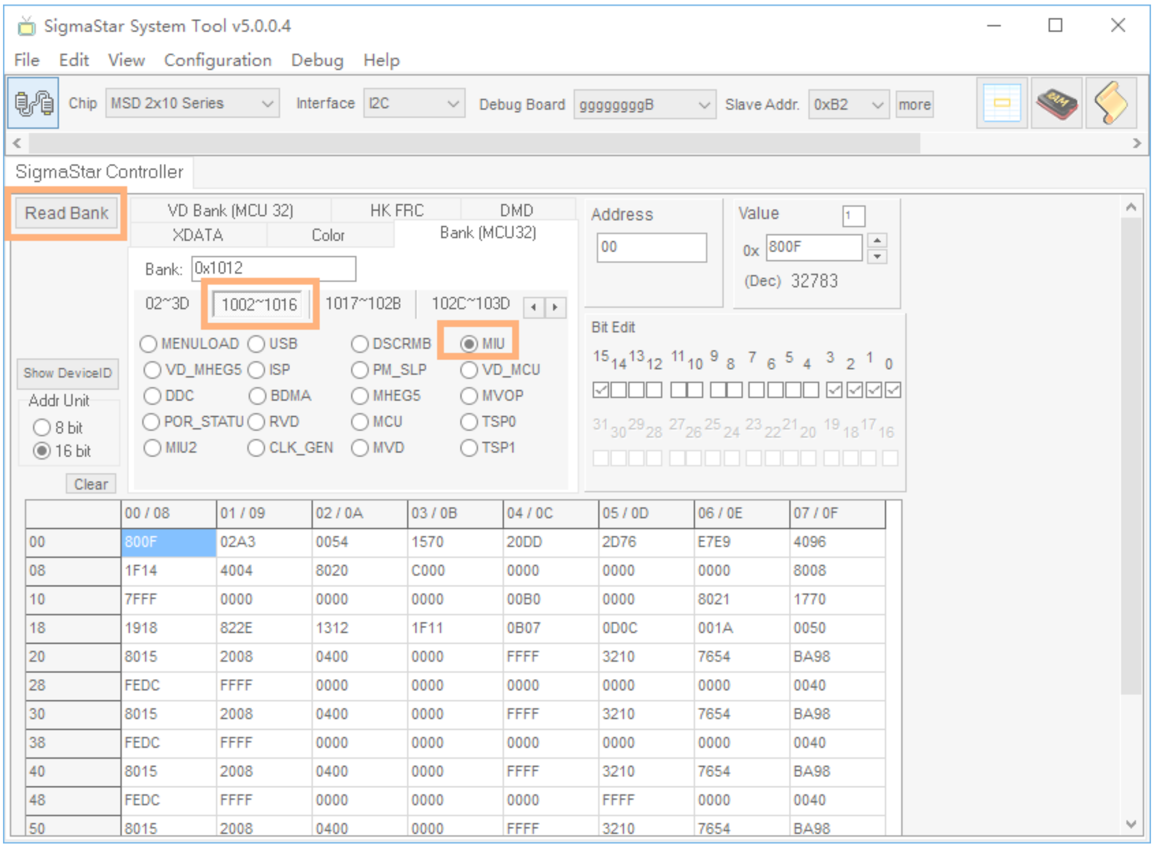


图 1-5

如图1-6所示：Bank1012 偏移地址1B 16bit的值是0x1F11

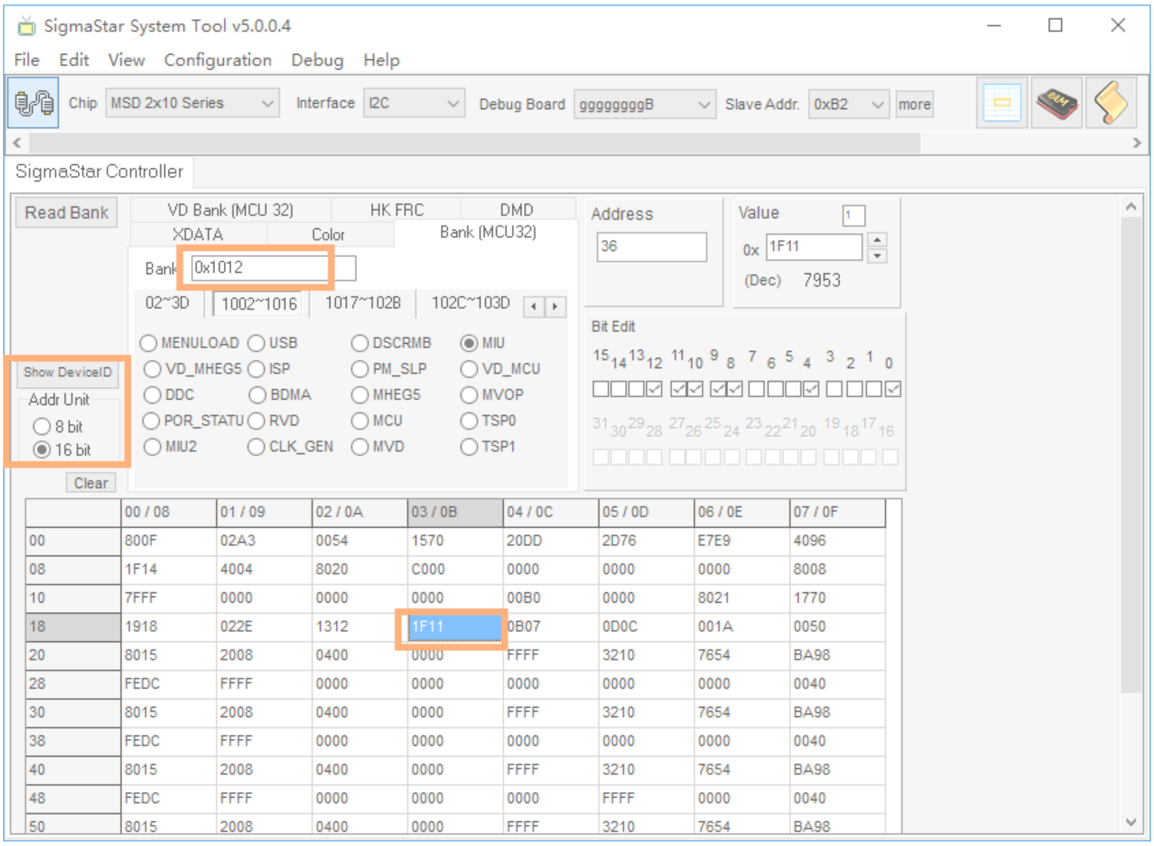


图 1-6

如图1-7所示：例如Bank1012 偏移地址36 8bit的值是0x1F 偏移地址35 8bit的值是0x11 8bit转换成16bit 是36/2=1B（16进制）

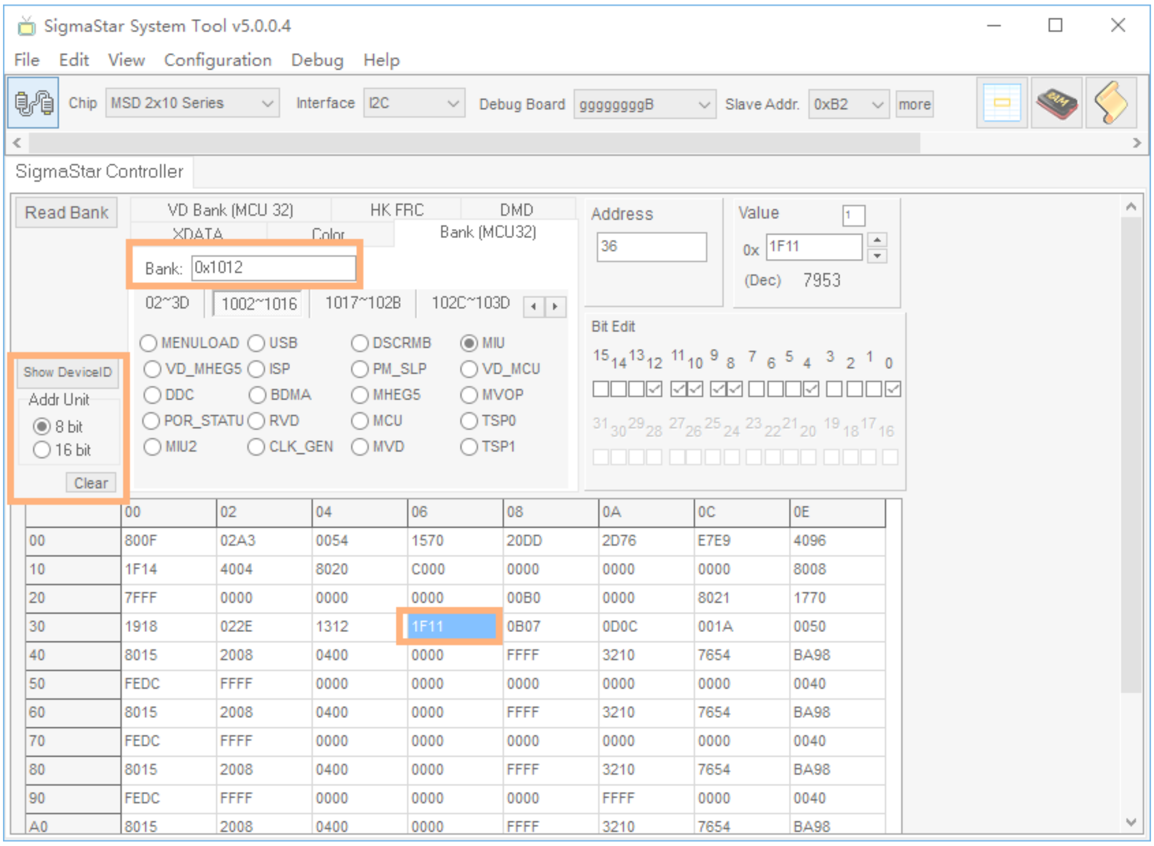


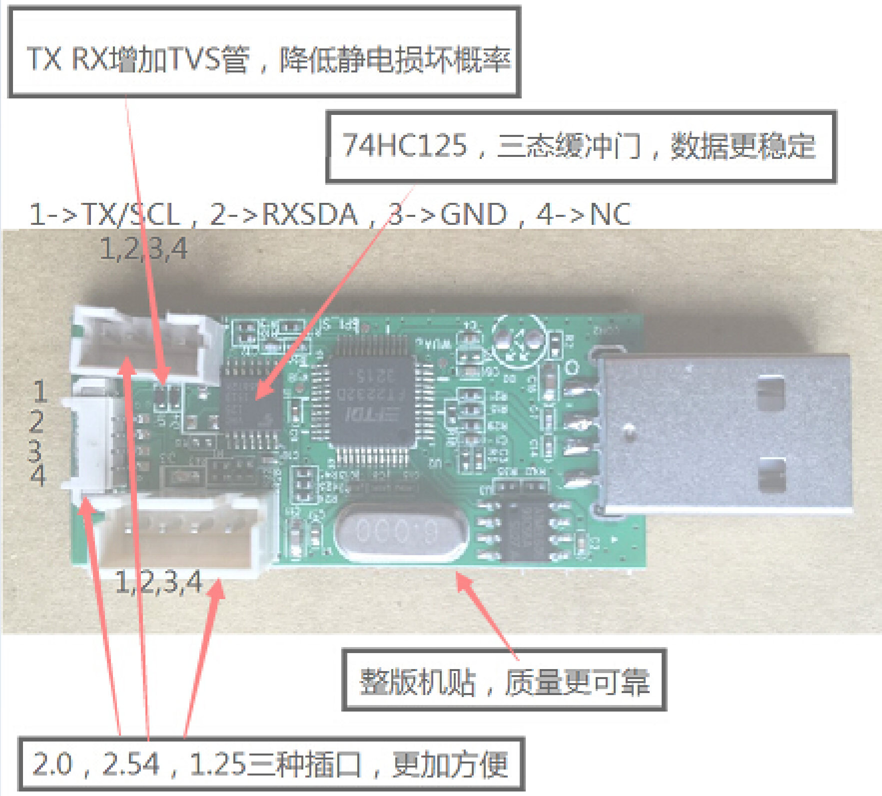
图 1-7

2. Sstar flash tool

2.1. 烧录硬件环境准备

2.1.1. Debug tool 硬件串口工具

此款工具专用于SigmaStar芯片的uboot烧录以及芯片Sstar tool寄存器访问。



2.1.2. 硬件连接框图

硬件单板		Debug Tool
PIN1 NC		PIN4 NC
PIN2 GND	↔	PIN3 GND
PIN3 RX(SDA)	←	PIN2 TX(SDA)
PIN4 TX(SCL)	→	PIN1 RX(SCL)

SigmaStar 芯片在单板顺序是NC GND RX TX

2.1.3. SPI-NOR Flash空片烧录

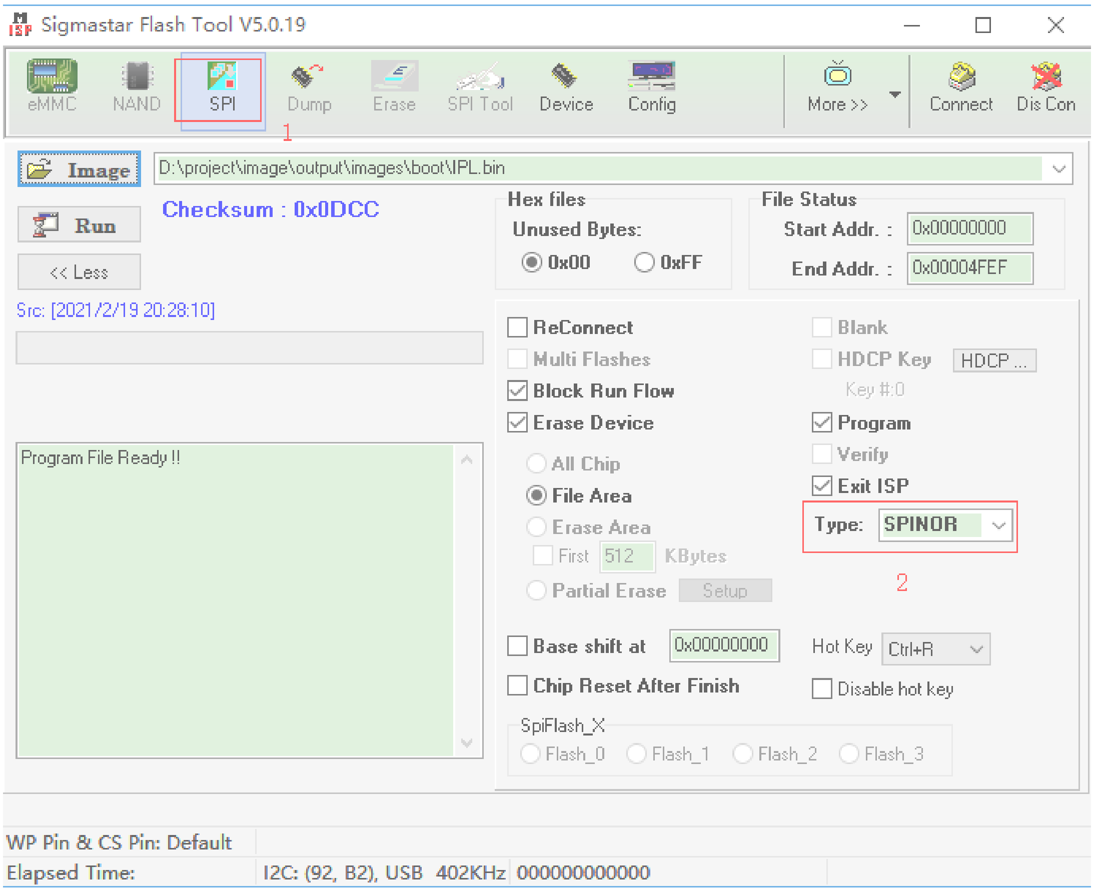
适用于空机烧录，或者uboot已经损坏导致无法通过uboot升级的场合。

SPI Norflash烧录的offset如下：

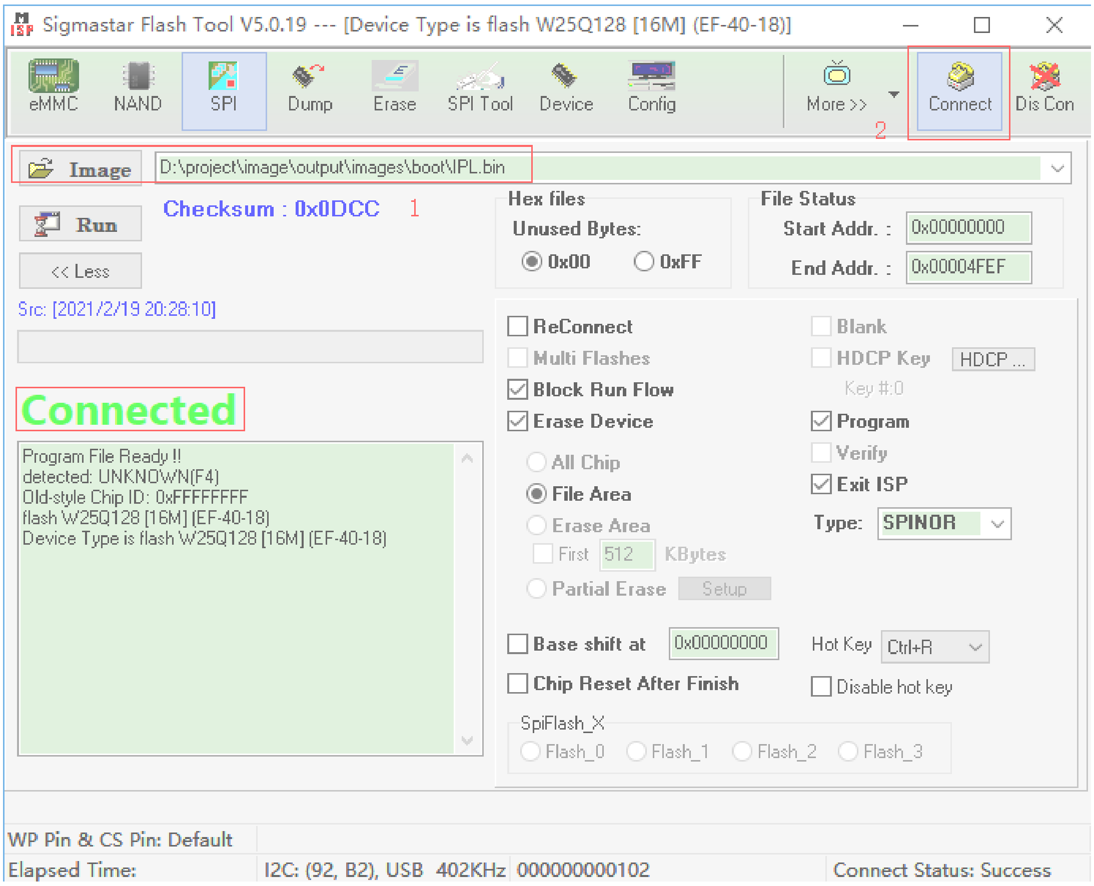
表 2-1

name	offset	path
IPL.bin	0x0000	project\image\output\i-mages\IPL.bin
IPL_CUST.bin	0x10000	project\image\output\images\IPL_CUST.bin
MXP_SF.bin	0x20000	project\image\output\images\MXP_SF.bin
u-boot.xz.img.bin	0x30000	project\image\output\images\u-boot.xz.img.bin

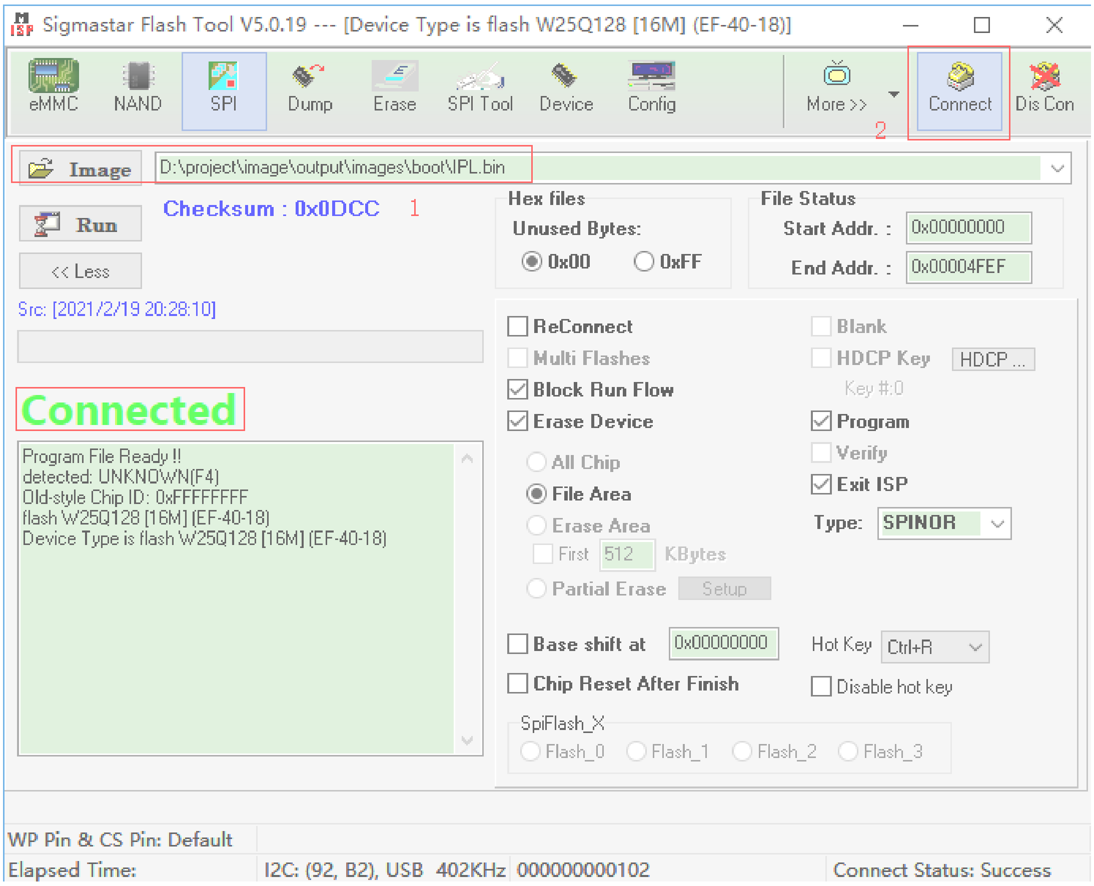
- Step 1:使用debug tool 连接板子，关闭串口终端，执行Flash Tool工具，板子上电。
- Step 2:选择 `SPI` tab，点击 `More` 并且选择类型为 `SPINOR` ；



- Step 3:加载烧录文件并点击 `Connect`

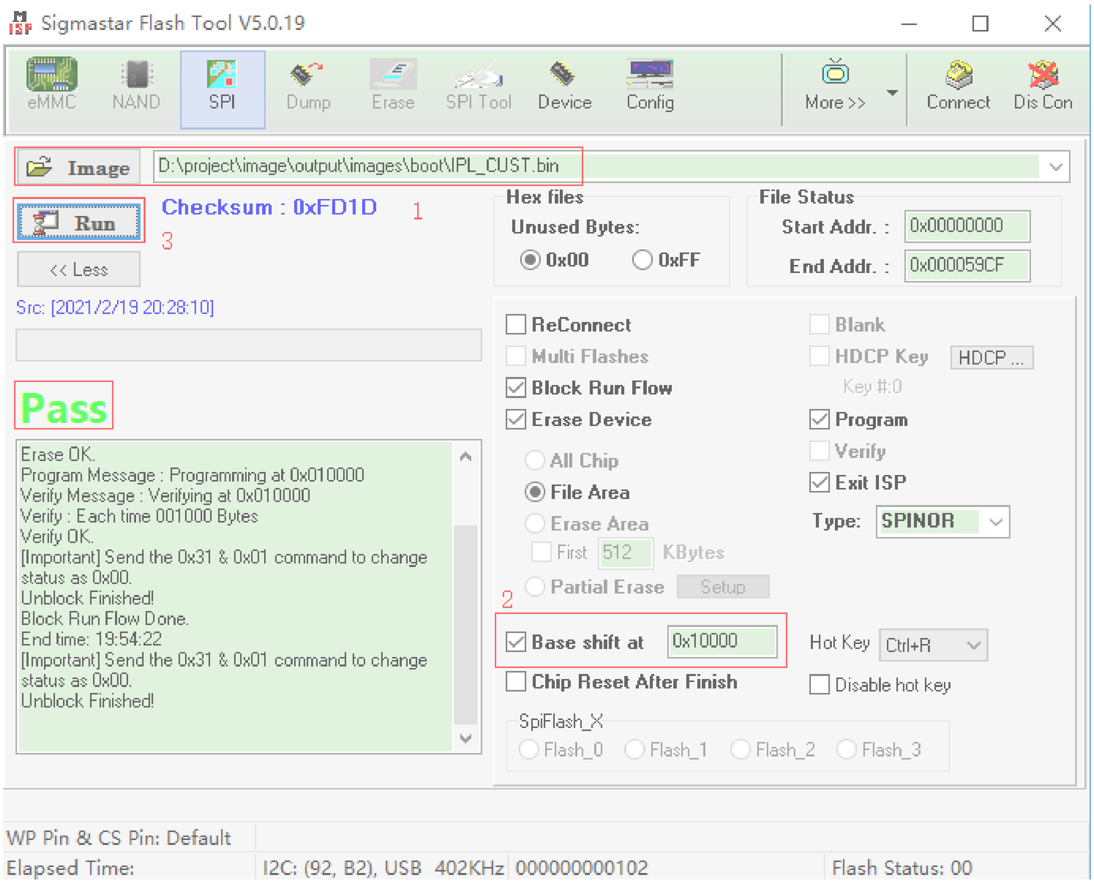


- Step 4:加载 `imageIPL.bin` ,点击 `Run` ；



- Step 5: 加载 image “IPL_CUST.bin”,设置 `Base shift` at 0x10000。

  注:可能随着版本变化，Base shift的地址以表2-1数据为准。



- Step 6: 加载image“MXP_SF.bin”, 设置“Base shift”at 0x20000。
- Step 7: 加载image “u-boot.xz.img.bin” , 设置“Base shift”at 0x30000。
- Step 9:重启板子即可

2.1.4. SPI-NAND FLASH空片烧录

适用于空机烧录，或者uboot已经损坏导致无法通过uboot升级的场合。

烧录SPINAND的方法和SPINOR一样，只是地址以及文件稍有差别。

表 2-2

Name	Erase Device	offset	Path
GCIS.bin	All Chip	0x00000	project/I2m/image
IPL.bin	File Area	0x140000	project/I2m/image
IPL_CUST.bin	File Area	0x200000	project/I2m/image
u-boot_spinand.xz.img.bin	File Area	0x2C0000	project/I2m/image

- Step 1:使用debug tool 连接板子，关闭串口终端，板子上电。
- Step 2:打开Flash Tool工具，如下图所示；

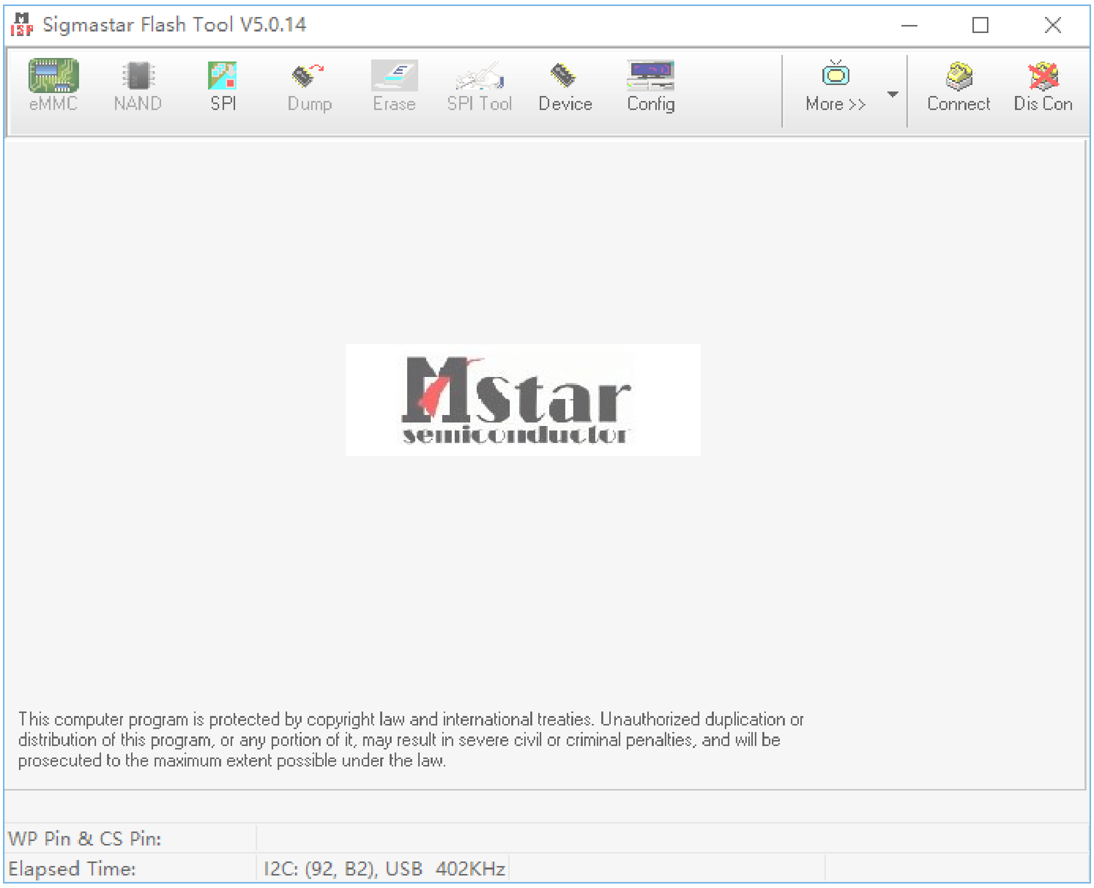


图 2-7

- Step 3:选择SPI按钮，出现如下图界面

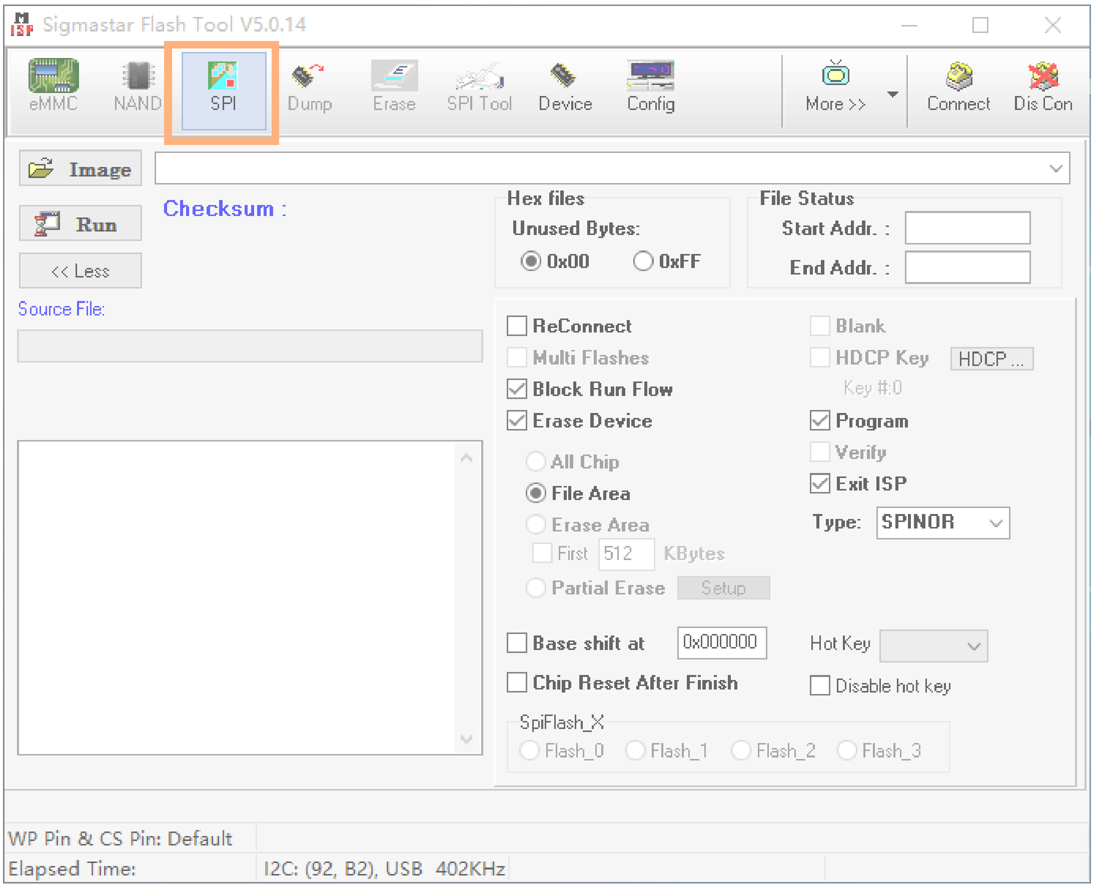


图 2-8

- Step 4:在Type按钮，选择SPINAND，连接Connect按钮，出现如下图界面，并点击确定。

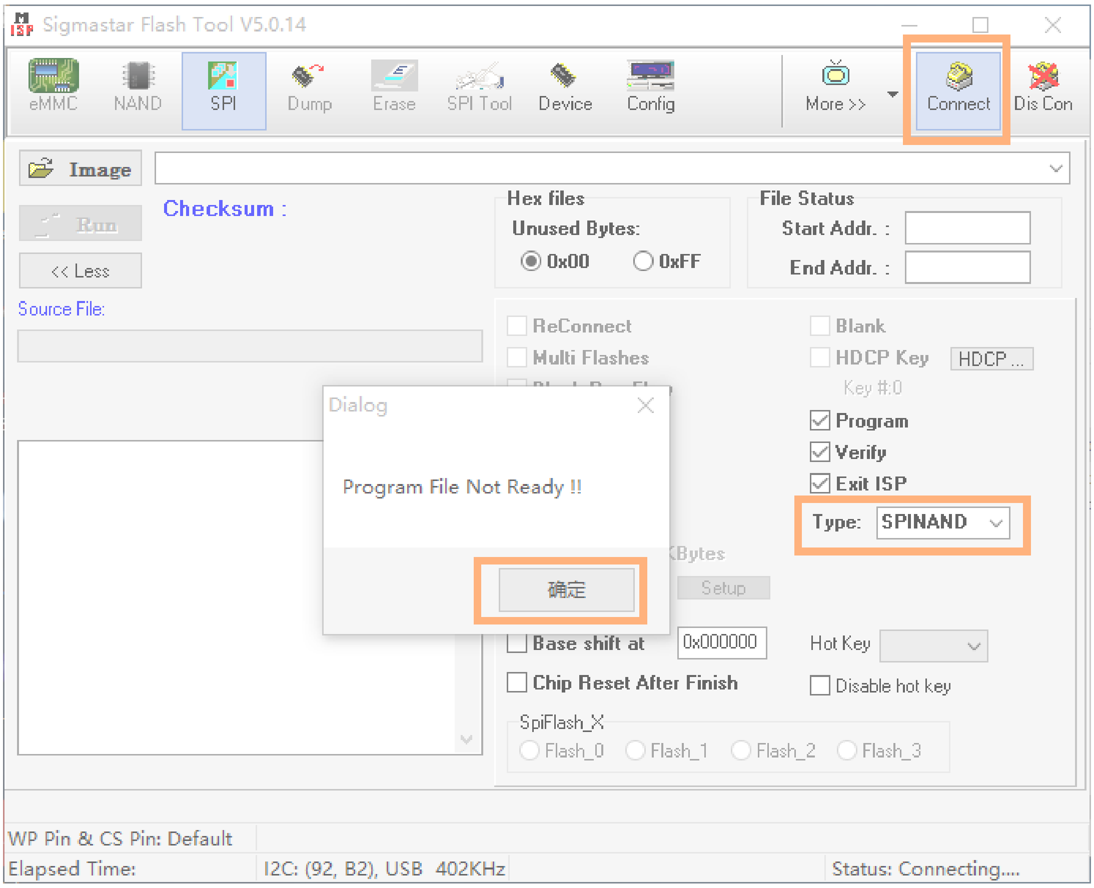


图 2-9

- Step 5:连接成功时，出现如下图16界面，显示对应的Flash型号，表示连接成功。

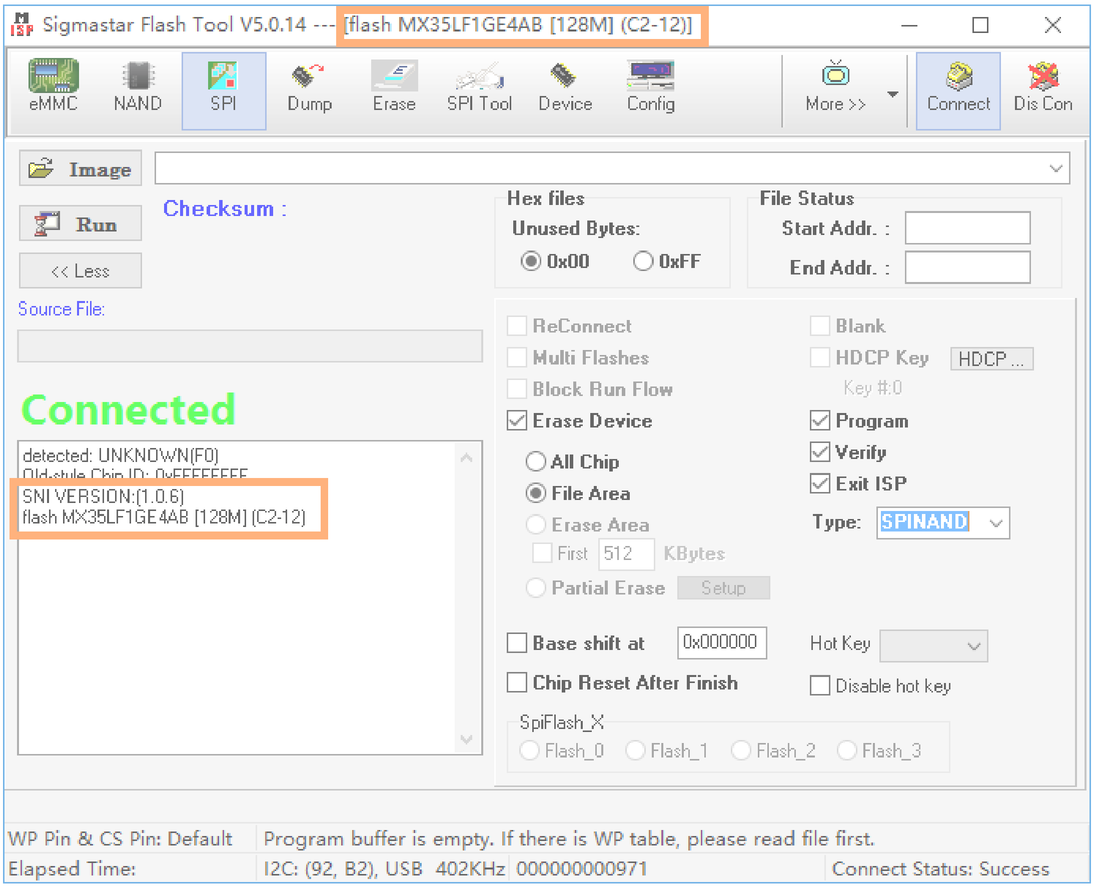


图 2-10

- Step 6:加载image“ GCIS.bin”,点击“Run”；

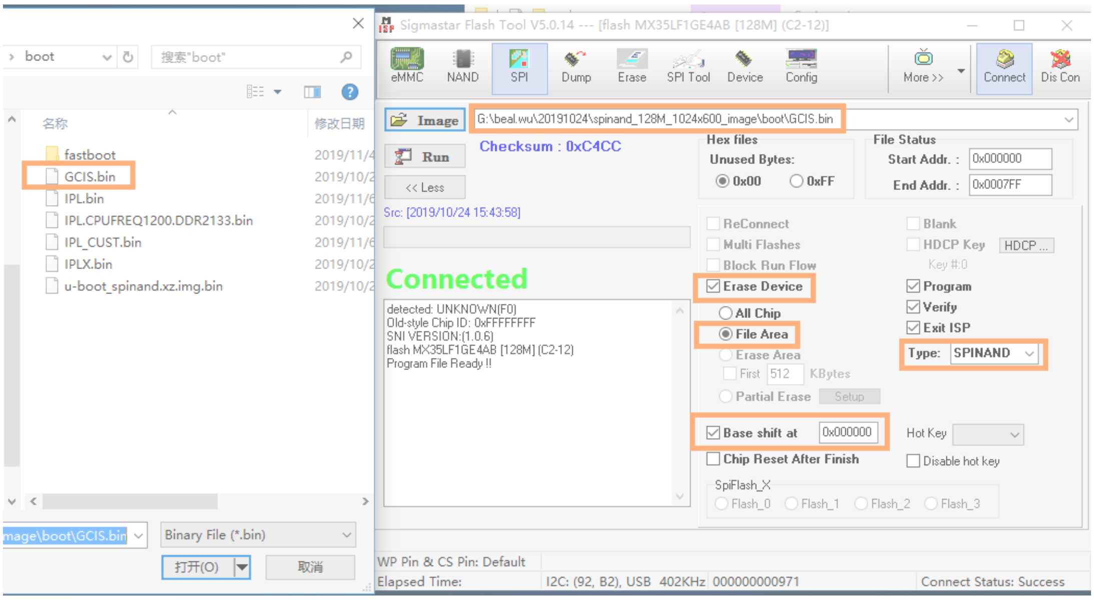


图 2-11

- Step 7: 加载 image “IPL.bin”,设置“Base shift”at 0x140000。

注:可能随着版本变化，Base shift的地址以表2-2数据为准。

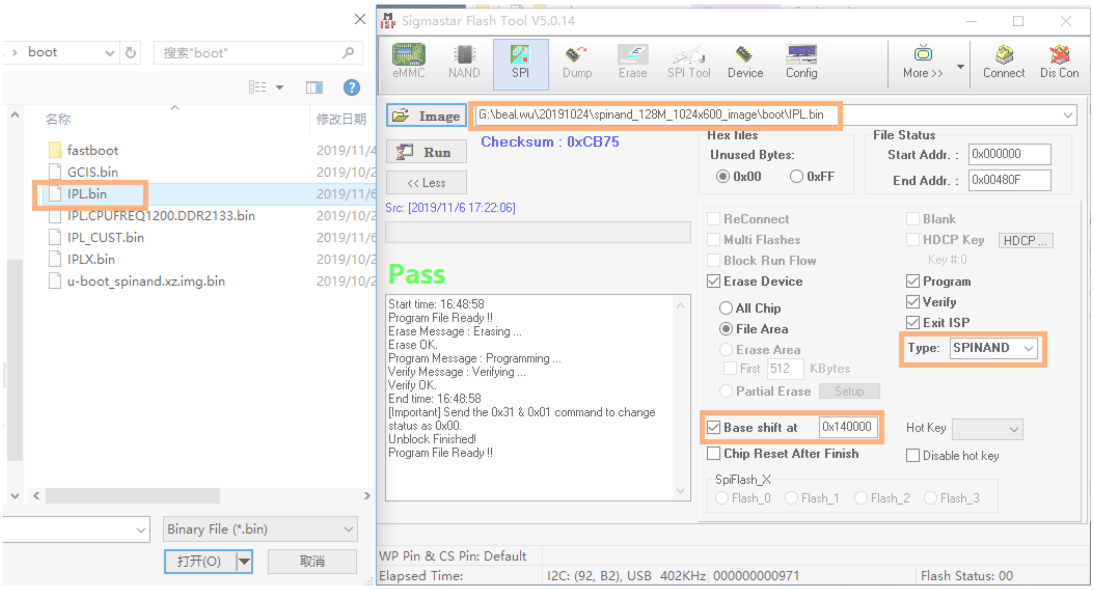


图 2-12

- Step 6: 加载image“IPL_CUST.bin”, 设置“Base shift”at 0x200000。

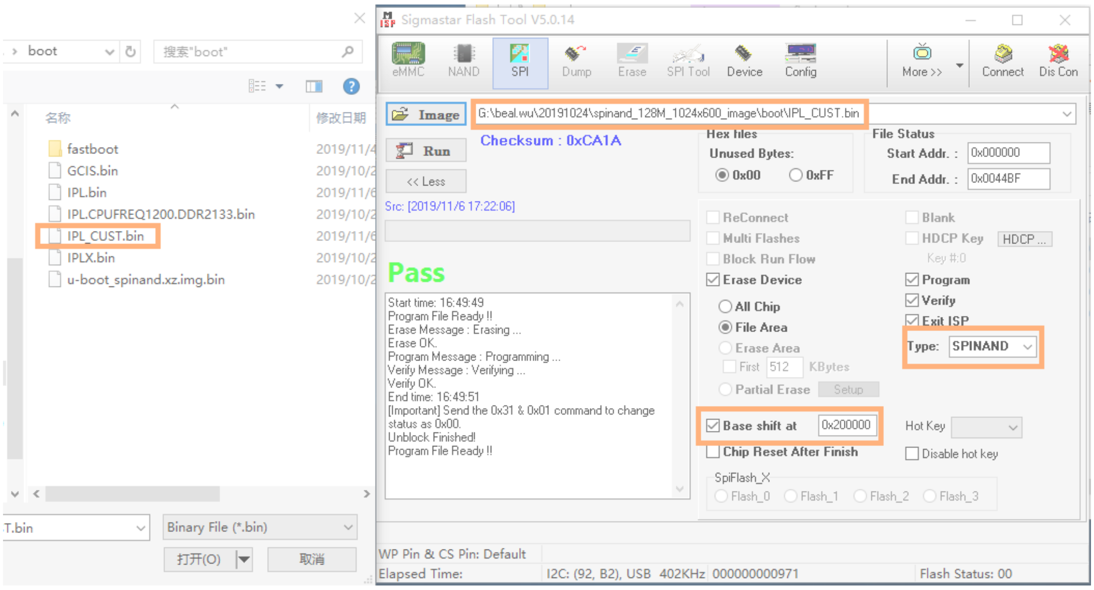


图 2-13

- Step 7: 加载image “u-boot_spinand.xz.img.bin” , 设置“Base shift”at 0x2c0000。

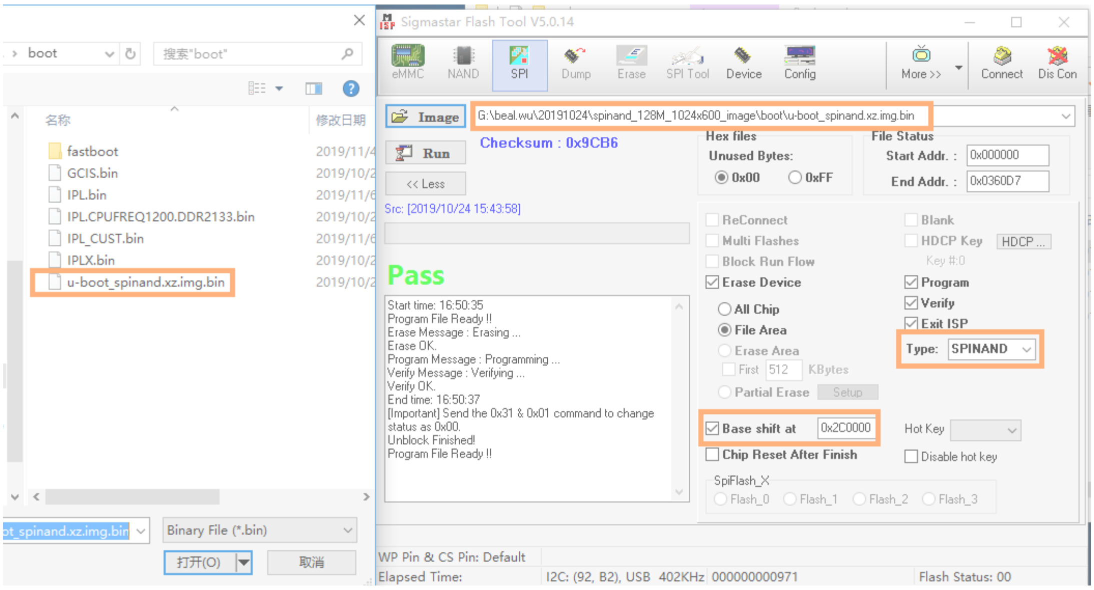


图 2-14

- Step 9:重启板子即可

2.1.5. Image烧录方法

在使用Flash tool工具烧录完uboot后，有三种方法可以烧录镜像文件，可以通过USB、SD卡、网络进行烧录。不同的芯片支持不同的烧录方法，这里重点介绍网络烧录，也是目前软件开发最常见的烧录方法。

使用网络烧录，需要准备PC机、交换机、开发板，网络拓扑如下：

图 2-15

- step1：确保网络连接正常，设置开发板IP地址和服务器IP地址，然后保存环境变量。

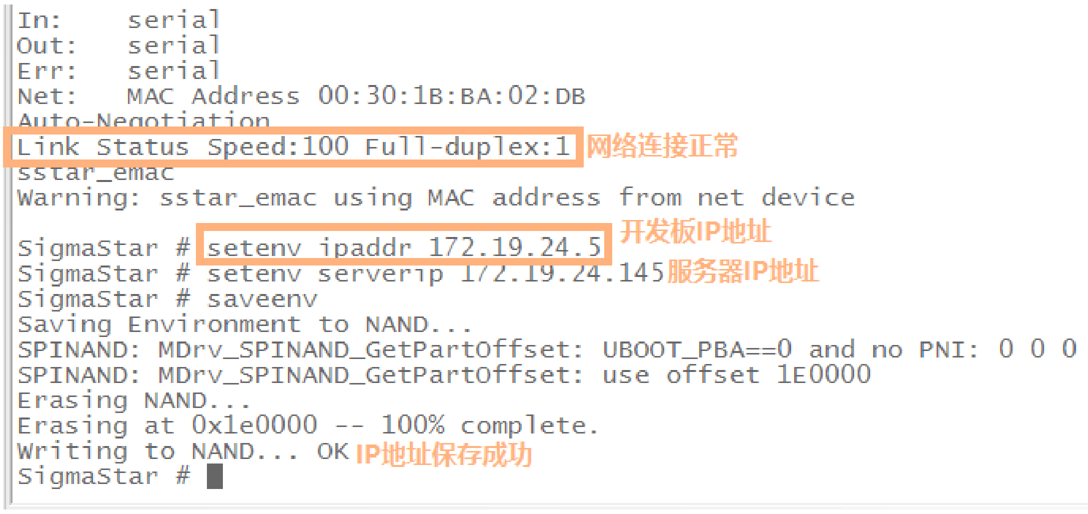


图 2-16

- step2：使用TFTP软件指定Image的烧录路径，确保服务器IP地址配置正确，如下图所示，板子Serverip是 `172.19.24.149`.

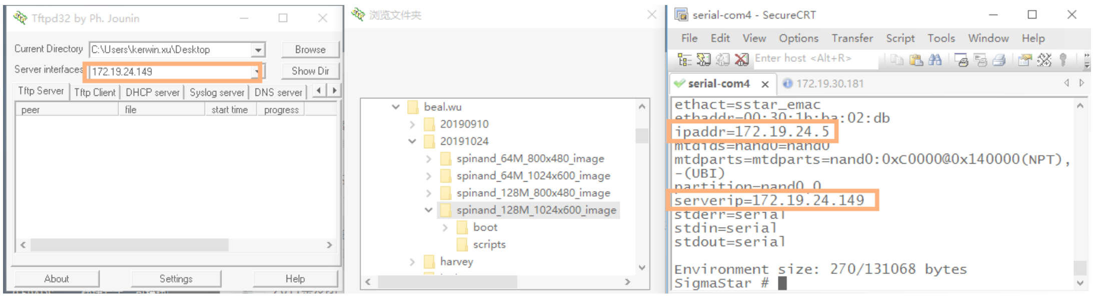


图 2-17

- step3：敲入“estar”命令，即可正常烧录。

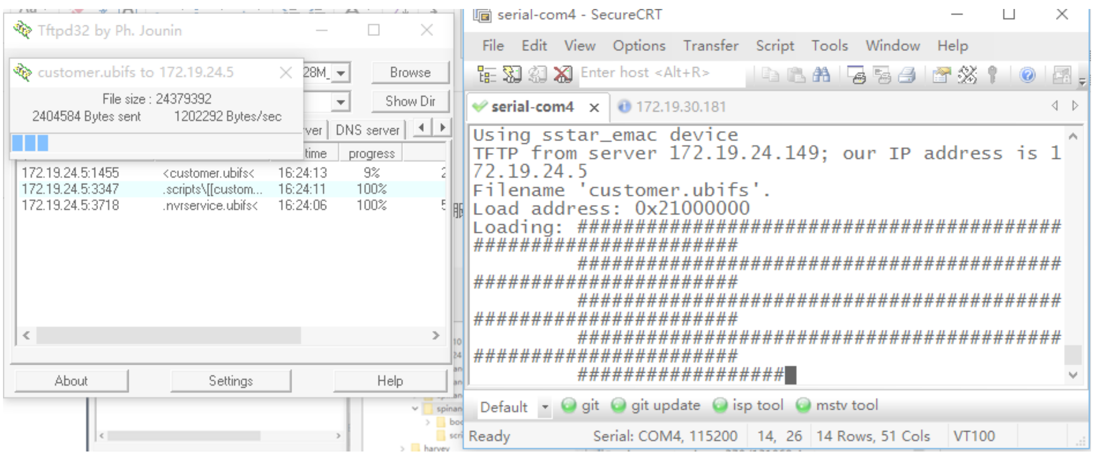


图 2-18

2.1.6. 如何擦除Flash

有时候flash中有程序，需要使用Flash tool软件工具擦除。如下图所示，选择正确的Flash类型，然后选择“All Chip”，并指定一个 `GCIS.bin` 文件，点击“Run”按钮，即可擦除Flash。

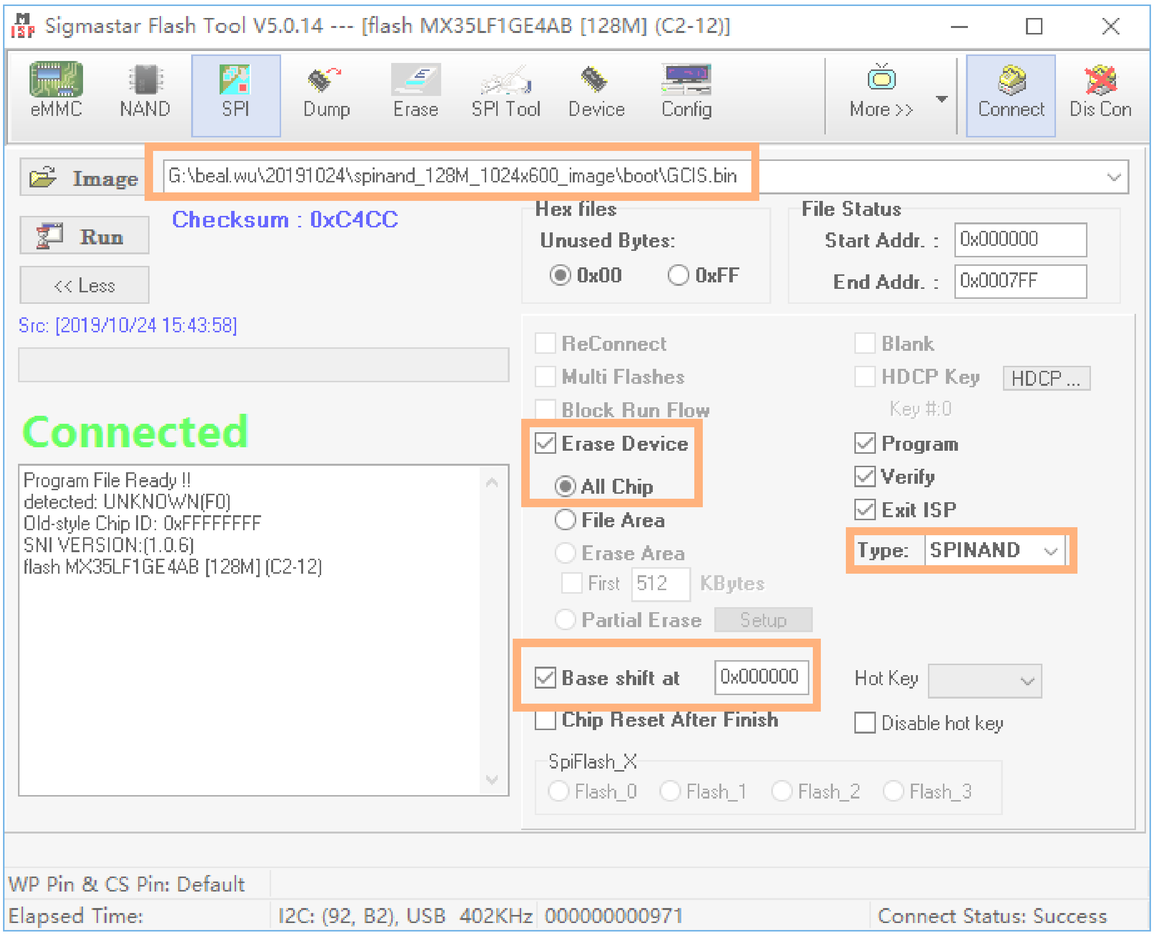


图 2-19

2.2. Sstar Flash Tool 烧录错误

2.2.1. 硬件连接错误

使用硬件工具Debug tool和芯片连接，需要注意串口的接线顺序，SigmaStar芯片调试串口Pin是 `PM_UART_RX/PM_UART_TX` ，接口顺序是NC GND RX TX，注意Debug Tool PCBA丝印命名，接法是RX接RX，TX接TX，GND接GND。

2.2.2. 串口工具选择错误

The empty chip programming does not use the debug tool hardware tool, but uses the common serial port tool, as shown in the figure below. The SigmaStar chip debugging serial port has the I2C function. When programming and accessing the register, the I2C function is used. The ordinary serial port printing tool does not have this function, so the hardware tool mentioned in Figure 1-1 above needs to be used.

Figure 2-20

2.2.3. Wrong choice of Flash type

During the SPInand programming process, the corresponding Flash type is not selected. For example, when programming SPINAND, SPINOR is selected. As shown in the figure below, if the wrong Flash type is selected, the chip displays connected, but the connection is not successful.

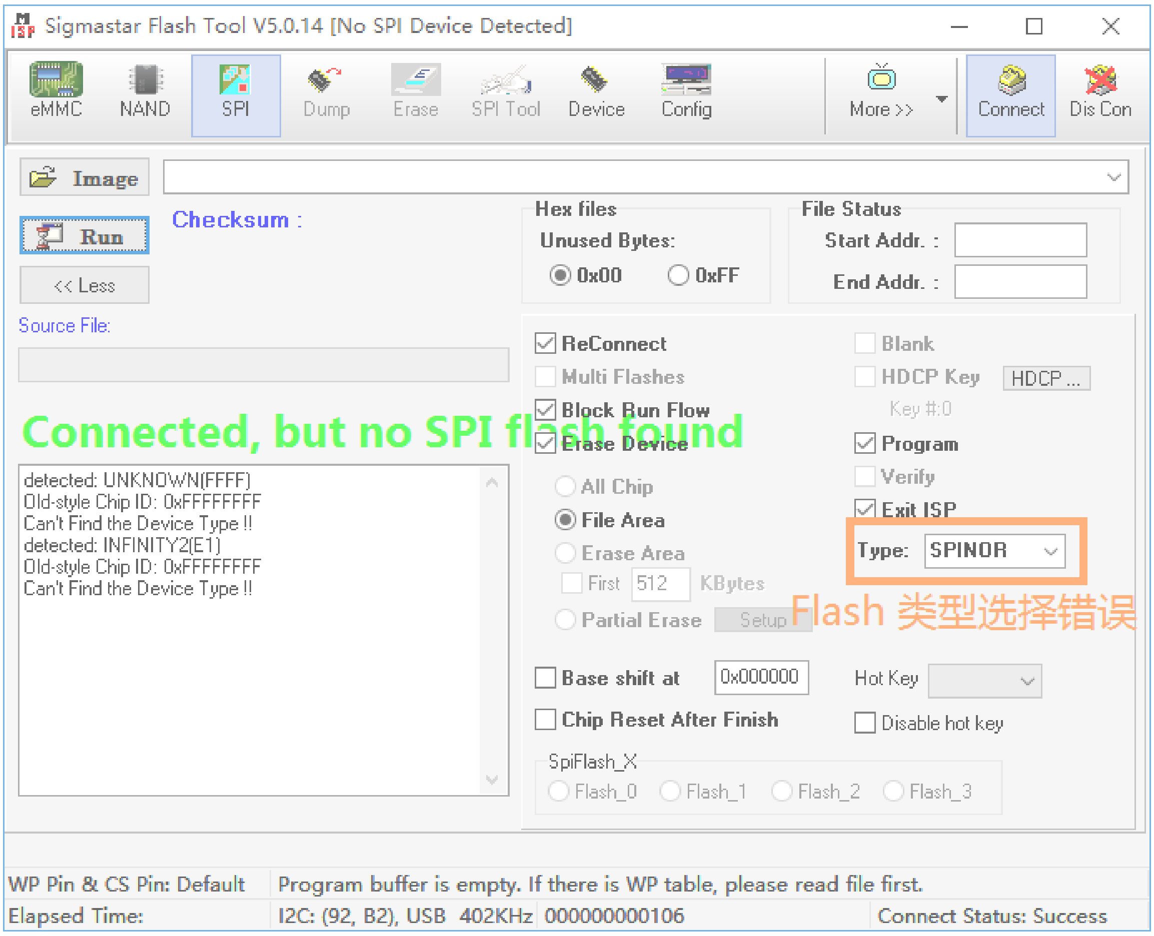


Figure 2-21

2.2.4. Flash tool file is missing

The Flash type is correctly selected, as shown in the figure below, but the SPINAND model is still not found. It may be that the file is missing `SPINANDINFO.sni` . It contains the SPINAND manufacturer, ID and other information.

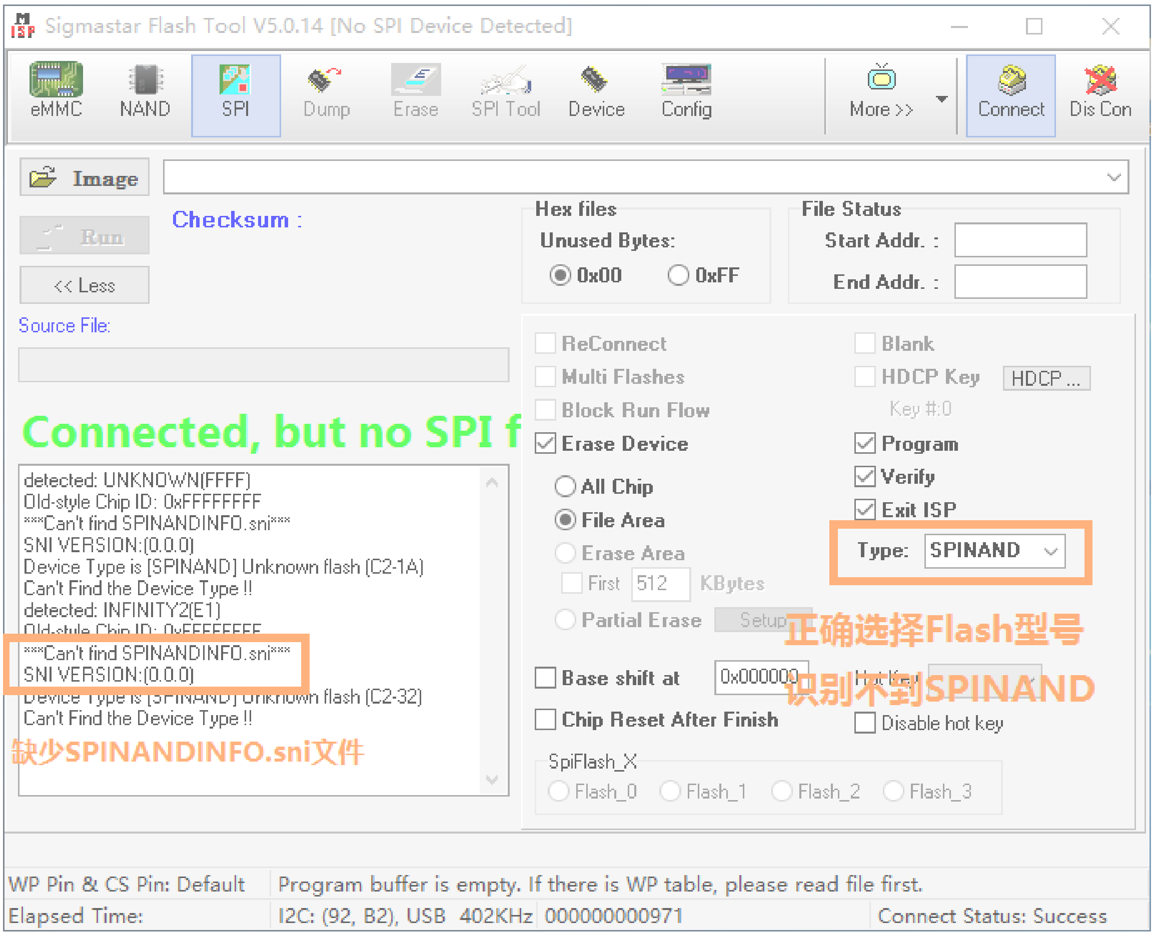


Figure 2-22

2.2.5. Serial terminal is not closed

When the Flash tool software is opened, the serial terminal software of the computer is not closed, so that the Flash tool cannot recognize the Flash.

Figure 2-23